

Practical-1

Aim

To perform Linux System Administration and File Management operations for the Marketing Analytics team using Linux commands, including directory creation, file management, searching, backup, and system monitoring.

Problem Definition:

Scenario: You have recently joined DataAnalytics Inc. as a Linux System Administrator. The company runs its analytics platform on Linux servers Where project data, reports, scripts, and user accounts must be managed efficiently. Your manager has assigned you the responsibility of setting up the project environment, managing files and directories, monitoring system resources, securing data and creating backups for the Marketing Analytics team.

Objectives

- Understand Linux command-line operations.
- Create and manage directories and files.
- Perform searching using Linux commands.
- Manage file backup and movement.
- Practice file deletion and directory management.
- Learn basic Linux system administration.

Tasks:

1. Identify the current user.



```
gitanshu@Parth:~$ whoami
gitanshu
```

2. Find the current working directory.

```
gitanshu@Parth:~$ pwd
/home/gitanshu
```

3. Display system information.

```
gitanshu@Parth:~$ uname -a
Linux Parth 5.10.16.3-microsoft-standard-WSL2 #1 SMP Fri Apr 2 22:23:49 UTC 2021 x86_64 x86_64 x86_64 GNU/Linux
```

4. Check system uptime and date.

```
gitanshu@Parth:~$ uptime
06:36:14 up 1:39, 0 user, load average: 0.03, 0.03, 0.00
gitanshu@Parth:~$ date
Sat Aug 1 06:36:25 UTC 2026
gitanshu@Parth:~$
```

5. Create a directory named MarketingAnalytics.

```
gitanshu@Parth:~$ mkdir MarketingAnalytics
```

6. Create subdirectories: RawData, ProcessedData, Reports.

```
gitanshu@Parth:~$ mkdir MarketingAnalytics/{RawData,ProcessedData,Reports}
```

7. Create data and report files.

```
gitanshu@Parth:~$ touch MarketingAnalytics/RawData/customer_data.txt
touch MarketingAnalytics/Reports/report1.txtgitanshu@Parth:~$ touch MarketingAnalytics/Reports/report1.txt
```

8. Add sample content.

```
gitanshu@Parth:~$ echo "CustomerID,Name,City" > MarketingAnalytics/RawData/customer_data.txt
```

9. Display file contents.

```
gitanshu@Parth:~$ cat MarketingAnalytics/RawData/customer_data.txt
CustomerID,Name,City
```

10. List all files and directories.

```
gitanshu@Parth:~$ ls -R MarketingAnalytics
MarketingAnalytics:
ProcessedData  RawData  Reports

MarketingAnalytics/ProcessedData:

MarketingAnalytics/RawData:
customer_data.txt  customer_data.txt.save

MarketingAnalytics/Reports:
report1.txt
```

11. Search for report files.

```
gitanshu@Parth:~$ find MarketingAnalytics -name "customer_data.txt"
MarketingAnalytics/RawData/customer_data.txt
```

12. Search specific content inside files.

```
gitanshu@Parth:~$ grep "Raj" MarketingAnalytics/RawData/customer_data.txt
```

13. Copy report files to Backup directory.

```
gitanshu@Parth:~$ mkdir Backup
```

14. Rename files.

```
gitanshu@Parth:~$ mv MarketingAnalytics/Reports/report1.txt MarketingAnalytics/Reports/final_report.txt
```

15. Move files between directories.

```
gitanshu@Parth:~$ mv MarketingAnalytics/RawData/customer_data.txt MarketingAnalytics/ProcessedData/
```

16. Delete a file.

```
gitanshu@Parth:~$ rm MarketingAnalytics/ProcessedData/customer_data.txt
```

17. Delete an empty directory.

```
gitanshu@Parth:~$ rmdir MarketingAnalytics/ProcessedData
```

Key Questions / Analysis:

Q1. What is the difference between absolute and relative paths?

Absolute Path

- Starts from the root directory (/).
- Independent of the current directory.
- Example:

/home/gitanshu/MarketingAnalytics/Reports/report1.txt

Relative Path

- Starts from the current working directory.
- Depends on the current location.
- Example:

Reports/report1.txt

Q2. Why are Linux permissions important for file security?

Linux permissions protect files from unauthorized access by controlling who can read, write, or execute files. They improve system security and prevent accidental modification or deletion of important data.

Q3. What is the purpose of the grep command?

The **grep** command searches for specific text or patterns inside files. It is commonly used for log analysis, searching keywords, and filtering output.

Example:

```
grep "Marketing" Reports/*.txt
```

Q4. How does Linux support efficient file management?

Linux provides powerful commands such as **mkdir**, **cp**, **mv**, **rm**, **find**, **grep**, **ls**, and supports permissions, ownership, scripting, and automation, making file management efficient.

Q5. Why are backups important in enterprise environments?

Backups protect important business data from accidental deletion, hardware failure, cyberattacks, and disasters. They help organizations quickly recover data and maintain business continuity.

Supplementary Problems

1. Create a shell script to automate project directory creation.

```
#!/bin/bash
```

```
mkdir -p MarketingAnalytics/{RawData,ProcessedData,Reports,Backup}
```

```
touch MarketingAnalytics/RawData/customers.csv
```

```
touch MarketingAnalytics/ProcessedData/clean_data.csv
```

```
touch MarketingAnalytics/Reports/report1.txt
```

```
echo "Project directories created successfully."
```

```
gitanshu@Parth:~$ chmod +x setup.sh
gitanshu@Parth:~$ ./setup.sh
Project directories created successfully.
```

2. Find the top 10 largest files in the system.

```
find / -type f -exec du -h {} + 2>/dev/null | sort -rh | head -10
```

3. Schedule a backup task using cron.

```
gitanshu@Parth:~$ crontab -e
crontab: installing new crontab
gitanshu@Parth:~$ crontab -l
0 18 * * * cp -r ~/MarketingAnalytics ~/MarketingBackup
gitanshu@Parth:~$
```

```
gitanshu@Parth: ~
GNU nano 7.2 /tmp/crontab.zP731T/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow  command
0 18 * * * cp -r ~/MarketingAnalytics ~/MarketingBackup

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line  ^E Redo       ^G Copy
```

4. Monitor CPU utilization continuously using top/htop.

```

gitanshu@Parth:~$ sudo apt install htop
htop
[sudo] password for gitanshu:
Sorry, try again.
[sudo] password for gitanshu:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libglapi-mesa
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libnl-3-200 libnl-genl-3-200
Suggested packages:
  lm-sensors
The following NEW packages will be installed:
  htop libnl-3-200 libnl-genl-3-200
0 upgraded, 3 newly installed, 0 to remove and 95 not upgraded.
Need to get 238 kB of archives.
After this operation, 665 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-3-200 amd64 3.7.0-0.3build1.1 [55.7 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnl-genl-3-200 amd64 3.7.0-0.3build1.1 [12.2 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 htop amd64 3.3.0-4build1 [171 kB]
Fetched 238 kB in 16s (14.6 kB/s)
Selecting previously unselected package libnl-3-200:amd64.
(Reading database ... 51746 files and directories currently installed.)
Preparing to unpack .../libnl-3-200_3.7.0-0.3build1.1_amd64.deb ...

```

```

0[|||||] 15.7% Tasks: 30, 20 thr, 0 kthr; 1 running
1[||] 3.4% Load average: 1.23 0.98 0.49
2[||] 6.1% Uptime: 00:09:28
3[|||||] 8.7%
Mem[|||||] 330M/3.65G
Swp[|] 0K/1.00G

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
515 gitanshu 20 0 4480 2944 2320 D 1.3 0.1 0:06.91 find / -type f -exec du -h {} +
1 root 20 0 21804 13200 9792 S 0.0 0.3 0:01.20 /sbin/init
2 root 20 0 3180 2208 2072 S 0.0 0.1 0:00.05 /init
6 root 20 0 3196 2132 2008 S 0.0 0.1 0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8
7 root 20 0 3196 2132 2008 S 0.0 0.1 0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8
8 root 20 0 3180 2208 2072 S 0.0 0.1 0:00.00 /init
56 root 19 -1 66700 15920 14768 S 0.0 0.4 0:00.41 /usr/lib/systemd/systemd-journald
106 root 20 0 25284 6740 5220 S 0.0 0.2 0:00.20 /usr/lib/systemd/systemd-udev
147 systemd-re 20 0 21344 13288 11164 S 0.0 0.3 0:00.10 /usr/lib/systemd/systemd-resolved
148 systemd-ti 20 0 91036 8052 7084 S 0.0 0.2 0:00.09 /usr/lib/systemd/systemd-timesyncd
161 systemd-ti 20 0 91036 8052 7084 S 0.0 0.2 0:00.00 /usr/lib/systemd/systemd-timesyncd
165 root 20 0 4244 2712 2464 S 0.0 0.1 0:00.01 /usr/sbin/cron -f -P
166 messagebus 20 0 9640 5516 4784 S 0.0 0.1 0:00.14 @dbus-daemon --system --address=systemd: --nofork --n
183 root 20 0 17984 8952 7912 S 0.0 0.2 0:00.10 /usr/lib/systemd/systemd-logind
186 root 20 0 1714M 13168 11000 S 0.0 0.3 0:00.10 /usr/libexec/wsl-pro-service -vv
189 syslog 20 0 217M 6116 4720 S 0.0 0.2 0:00.05 /usr/sbin/rsyslogd -n -iNONE
206 root 20 0 3124 2040 1900 S 0.0 0.1 0:00.01 /sbin/agetty -o -p -- \u --noclear - linux
209 syslog 20 0 217M 6116 4720 S 0.0 0.2 0:00.01 /usr/sbin/rsyslogd -n -iNONE
210 syslog 20 0 217M 6116 4720 S 0.0 0.2 0:00.00 /usr/sbin/rsyslogd -n -iNONE
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice + F9Kill F10Quit

```

5. Generate a disk usage report for all project directories.

```

gitanshu@Parth:~$ du -sh MarketingAnalytics/*
4.0K MarketingAnalytics/Backup
4.0K MarketingAnalytics/ProcessedData
8.0K MarketingAnalytics/RawData
4.0K MarketingAnalytics/Reports
gitanshu@Parth:~$

```

Tools /Technology:

Ubuntu Linux / CentOS Linux

Linux Terminal

Bash Shell

Conclusion:

This practical demonstrated essential Linux system administration and file management skills. Operations such as creating directories, managing files, searching data, backing up project files, monitoring system resources, and automating tasks using shell scripts were successfully performed. These skills are fundamental for efficiently managing enterprise Linux environments used in data analytics projects.

Github Repo:

https://github.com/Gitanshu1504/24DIT070_BDA_Practicals_2026_SHP